Grade 7 Accelerated　　　Name__Answer Key_______
Unit 1　　　　　　　　　Date___________________
Lesson 5 Practice　　　　Period__________________

1. Solve the equation $4(0.5p+1.5)=12$ using your preferred method.

$$\frac{4(0.5p+1.5)}{4}=\frac{12}{4}$$
$$0.5p+1.5=3$$
$$\underline{-1.5\quad -1.5}$$
$$\frac{0.5p}{0.5}=\frac{1.5}{0.5}$$
$$p=3$$

2. Use substitution to verify your solution in question 1.

$$4(0.5(3)+1.5)=12$$
$$4(1.5+1.5)=12$$
$$4(3)=12$$
$$12=12$$

3. Solve the equation $\frac{5}{2}=\frac{1}{4}(z-8)$ using your preferred method.

$$4\cdot\frac{5}{2}=4\cdot\frac{1}{4}(z-8)$$
$$10=z-8$$
$$\underline{+8\quad +8}$$
$$18=z$$

4. Use substitution to verify your solution in question 3.

$$\frac{5}{2}=\frac{1}{4}(18-8)$$
$$\frac{5}{2}=\frac{1}{4}(10)$$
$$\frac{5}{2}=\frac{5}{2}$$

5. Solve the equation $18=\frac{3}{4}(36.4-n)$ using your preferred method.

$$\frac{4}{3}\cdot 18=\frac{4}{3}\cdot\frac{3}{4}(36.4-n)$$
$$24=36.4-n$$
$$\underline{-36.4-36.4}$$
$$\frac{-12.4}{-1}=\frac{-n}{-1}$$
$$12.4=n$$

6. Use substitution to verify your solution in question 5.

$$18=\frac{3}{4}(36.4-12.4)$$
$$18=\frac{3}{4}(24)$$
$$18=18$$

Josephine paid $\$650$ to rent a beachfront cabana at a vacation resort. For each guest that used the cabana a $\$19.40$ resort fee was added to the bill. If the bill was $\$882.80$, how many guests used the cabana?

7. What was the total amount of money spent on the cabana?

   $\$882.80$

8. Which cost depends on the number of people who use the cabana?

   $\$19.40$ resort fee per guest

9. Which cost is the same, regardless of how many visit the cabana?

   $\$650$ cabana rental fee

Sade was asked to solve the equation $\frac{2}{3}(b+2.5)=9.5$. Her work is shown.

$$\frac{2}{3}(b+2.5)=9.5$$
$$\frac{2}{3}\cdot b+\frac{2}{3}\cdot 2.5=9.5\cdot\frac{2}{3}$$
$$\frac{2}{3}b+\frac{5}{3}=\frac{19}{3}$$
$$\frac{2}{3}b+\frac{5}{3}-\frac{5}{3}=\frac{19}{3}-\frac{5}{3}$$
$$\frac{2}{3}b=\frac{14}{3}$$
$$\frac{3}{2}\cdot\frac{2}{3}b=\frac{3}{2}\cdot\frac{14}{3}$$
$$b=\frac{14}{2}=7$$

10. Explain where Sade made a mistake and find the correct solution.

    Sade multiplied both sides of the equation by $\frac{2}{3}$ when she should have distributed $\frac{2}{3}$ to $(b+2.5)$.